# Aditya Morolia

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## Education

**B.Tech in Computer Science and MS by Research in Computational Natural Sciences** — **IIIT Hyderabad**
*July 2017–June 2023* — *India*

**PhD, Computer Science** — **Centre for Quantum Technologies (CQT), National University of Singapore (NUS)**
*August 2024–Present* — *Singapore*

**Advisors:** Divesh Aggarwal
**Interests:** I work on randomized and quantum algorithms for optimization, cryptography, learning, and high-dimensional geometry. I particularly enjoy designing algorithms for computationally challenging problems, especially those involving large amounts of data, where randomness can be exploited to obtain efficient solutions.

## Publications

- Divesh Aggarwal, Aditya Morolia, and Noah Stephens-Davidowitz. Faster provable lattice sieving with spherical codes. Cryptology ePrint Archive, Paper 2026/2388, 2026 (eprint)
- Divesh Aggarwal, Rishav Gupta, Aditya Morolia, and Chuanqi Zhang. Mind the Gap? Not for SVP Hardness under ETH! In *ICALP 2026*, volume 374 of *LIPIcs*, pages 8:1–8:24, 2026 (arXiv; ICALP26)
- Joao F. Doriguello, George Giapitzakis, Alessandro Luongo, and Aditya Morolia. On the practicality of quantum sieving algorithms for the shortest vector problem. In *PQCrypto 2026*, volume 16491 of *LNCS*, pages 3–36, 2026 (eprint; QCTiP'25; PQCrypto'26)
- Shantanav Chakraborty, Aditya Morolia, and Anurudh Peduri. Quantum Regularized Least Squares. *Quantum*, 7:988, April 2023 (Quantum Journal)

## Experience

**Research Internship** — **CIFRA, Bocconi University, Milano**
*November 2025 - February 2026* — *Italy*
- Visiting Giulio Malavolta and Alon Rosen.
- Algorithms for Lattice Isomorphism; Cryptography for AI Safety

**Research Assistant** — **Centre for Quantum Technologies (CQT), National University of Singapore (NUS)**
*August 2023 - July 2024* — *Singapore*
- Paper: Quantum cryptanalysis of post-quantum lattice based schemes.

**Research Assistant** — **International Institute of Information Technology (IIIT) Hyderabad**
*May 2019 - June 2023* — *Hyderabad, India*
- Advisors: Shantanav Chakraborty, Indranil Chakrabarty
- MS Thesis: Quantum Algorithms for Regularized Least Squares.

**Mitacs Globalink Research Intern** — **University of Calgary**
*August 2021 - December 2021* — *Calgary, Canada*
- Advisor: Barry C Sanders
- Searching for Quantum Supremacy over classical Monte Carlo methods, using Quantum Annealing

**Engineering Analyst** — **Goldman Sachs**
*July 2022 - July 2023* — *GS Hyderabad, India*
- Worked on distributed database systems for financial trading in the SecDB team; focused on modularizing the monolithic C++ application. I was previously a summer analyst in the same team (2021).

**Technical Staff** — **AICrowd**
*April 2020–July 2020* — *India*
- Developed a tool to simulate user behaviour on the website to study ranking systems.

**Research Intern** — **Indian School of Business (ISB), Mohali**
*June 2019–August 2019* — *India*
- Analyzed basketball performance against pollsters' predictions with Dr. Siddharth Sharma.

## Selected Projects

- **Distributed MapReduce** (C++, Boost, MPI) — Worker-failure handling and load balancing. GitHub
- **C- - Compiler** (C++, Antlr) — Compiler for a toy language inspired by C and Decaf. GitHub
- **C Shell** (C, Linux) — Toy shell. GitHub
- **Graph Neural Network** (PyTorch, Cirq) — ParticleNet for particle data classification. GitHub
- **QSim** (TypeScript, CSS, SVG, Vite) — Offline circuit simulator, puzzles and synchronized displays. GitHub
- **Lattice Visualizer** (TypeScript, Canvas, Vite) — Lattices are beautiful mathematical objects. Let's try to see them! GitHub
- **Automated Bidding Agent** (Java) — Budget-constrained bidding for targeted ad space.
- **Molecular Dynamics Simulation** (Python) — MCMC with periodic boundaries and minimum-image convention.

## Teaching And Mentorship

**Teaching Assistant** — **NUS, Singapore**

- **Optimization for Machine Learning**, 2026. Instructors: Patrick Rebentrost and Maciej Obremski.
- **Quantum Computing**, 2026. Instructor: Divesh Aggarwal.
- **Advanced Algorithms (Lattices, Integer Programming)**. Instructor: Divesh Aggarwal. lattice lecture notes.

**Instructor, QCamp 2026** — **Centre for Quantum Technologies, Singapore**

- Taught complex numbers and linear algebra to pre-university students. Built interactive TypeScript/Canvas visualizations and companion slides to explain the mathematics underlying quantum computing. Interactive tools; Slides; Code

**Teaching Assistant** — **IIIT Hyderabad, India**

- **Linear Algebra**, 2020 (Undergraduate and Graduate)
- **Introduction to Quantum Information and Computation**, 2021 (Elective)
- **Automata Theory**, 2021 (Undergraduate)
- **Quantum Algorithms**, 2022 (Elective)

**Guest Lecture for Linear Algebra Course: Applications of Linear Algebra** · 2022 — **IIIT Hyderabad, India**

**Mentor, Singapore Quantum Youth Collective (SQYC)** · 2026 — **Singapore**

**Mentor, Women in STEM Research (WISR)** · 2023 — **Remote**

## Selected Talks

- **Expanding Frontiers of Cryptography: Assumptions & Applications, IIT Madras** — Faster Provable Lattice Sieving with Spherical Codes (Paper) — Nov 2026
- **School of Computing, NUS** — Learning Quantum Hamiltonians from Ground States — Oct 2026
- **School of Computing, NUS** — Quantum Monte Carlo Methods, Simplified — Oct 2026
- **NUS Group Seminar** — Is there a Johnson-Lindenstrauss lemma for the $\ell_1$ metric? — Oct 2026
- **NUS Group Seminar** — Pseudorandom Error-Correcting Codes — May 2026
- **NUS** — LoRA+: Efficient Low Rank Adaptation of Large Models (Slides) — Apr 2026
- **CQT, NUS** — Gaussian Boson Sampling (Slides) — Mar 2026
- **CQT CS Seminars** — Quantum Algorithms for Regularized Least Squares (Paper) — Nov 2023
- **Singapore QML Seminars** — Quantum Signal Processing and Applications to Linear Systems (Paper) — Nov 2023
- **Goldman Sachs, India Developers' Exchange** — Quantum Computing for Software Engineers — Mar 2023
- **IIIT Hyderabad Theory Group** — Quantum Approximate Counting — Aug 2020
  *Started the Theory Reading Group's initiative "Seminar Saturdays" with a this talk.*

## Awards and Honours

- National Quantum Scholarship Scheme (NQSS) award (National Quantum Office, Singapore) *2024*
- Mitacs Globalink Undergraduate Research Fellowship *2021*
- IIIT Hyderabad Dean's Merit List Award for Academic Performance (top 20% of cohort) *Spring 2021*
- IIIT Hyderabad Dean's Merit List Award for Academic Performance (top 20% of cohort) *Monsoon 2020*
- AICrowd DroneRL (Reinforcement Learning) Competition winner *2020*
- Qualified JEE (Mains) (Among top 0.1% candidates) *2017*
- Qualified JEE (Advanced) (Among top 0.3% candidates) *2017*